



EventOS – Complete Master Project

Theory, Architecture & Technical Blueprint



1. Executive Summary & Core Vision

EventOS is an enterprise-grade, multi-tenant operating system built explicitly for event planners, wedding agencies, concert producers, venue operators, and scenography directors.

The Problem It Solves

Traditional event management relies on fragmented, disconnected tools:

- Spreadsheets for budgeting & vendor payouts.
- Unstructured WhatsApp groups for client communication & crew dispatch.
- Paper/PDF proposals with manual invoicing & delayed payment collection.
- Disconnected photo galleries and missing run-of-show stage timing tools.

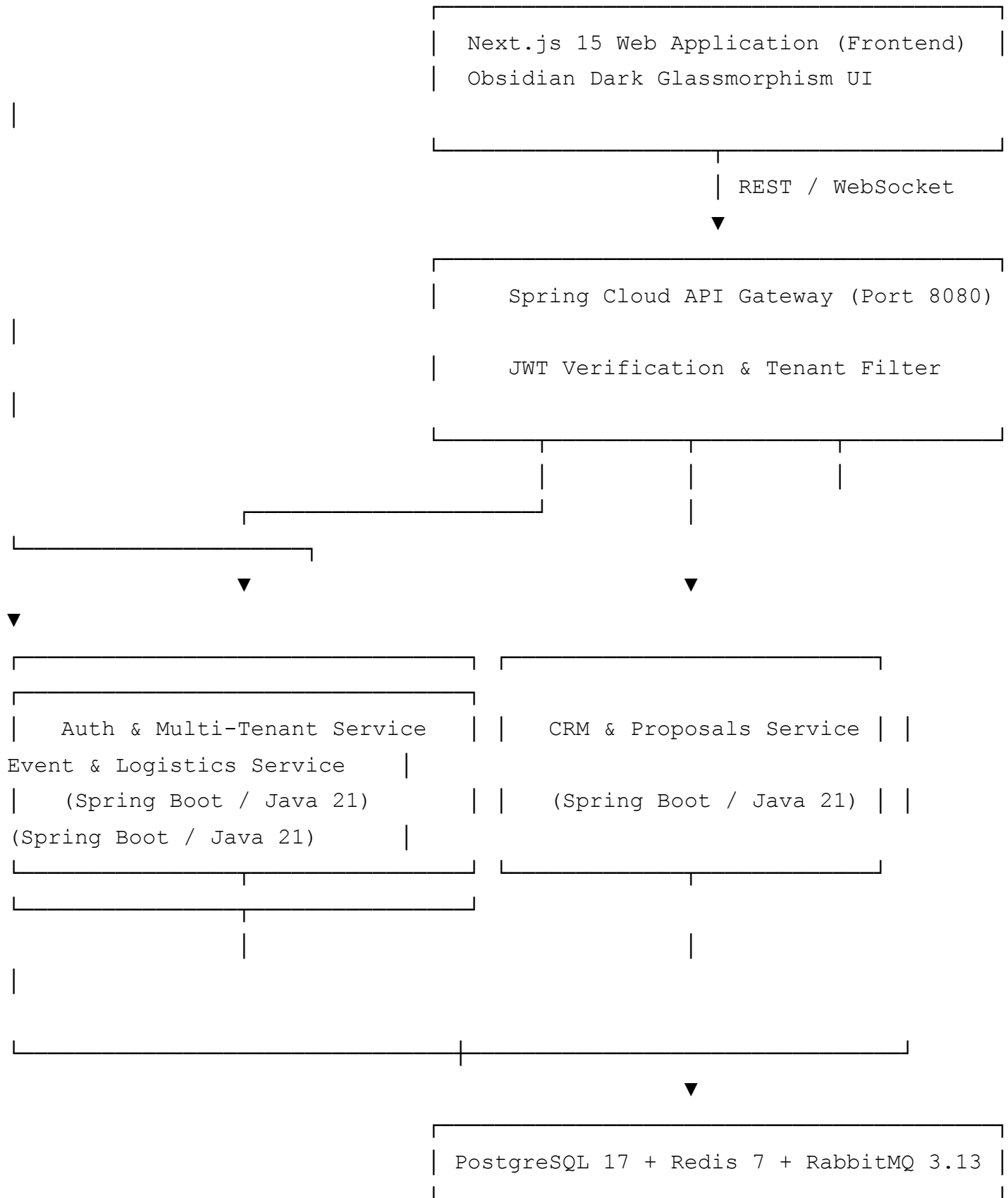
The EventOS Solution

EventOS replaces this fragmented chaos with a single, high-performance, real-time operating system that unifies:

1. **Multi-Tenant Agency Workspace:** Agencies manage multiple brand identities/clients with 1-second workspace switching.
 2. **Instant Quote & Contract Engine:** Dynamic cost estimation with 1-click PDF proposal exports.
 3. **Live Run-of-Show Stage Manager:** Real-time cue-by-cue sound, lighting, pyrotechnics, and crew dispatch timeline.
 4. **Financial Analytics & Margins Engine:** Real-time gross revenue, vendor expense breakdown, and net profit margin auditing (`EventFinancialAnalytics.tsx`).
 5. **Omnichannel Communication:** Integrated WhatsApp Business API 6-digit OTPs & automated booking alerts.
 6. **AI Event Assistant:** Automated event checklist, vendor recommendation, and contract generation.
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2. Architectural Blueprint & Technical Stack



Frontend Architecture (Web Layer)

- **Framework:** Next.js 15 (App Router, React 19, TypeScript).

- **Styling & Aesthetics:** Obsidian `#09090b` palette, Tailwind CSS, Framer Motion (60fps spring physics), Lenis smooth scroll, Lucide icon system.
- **State Management:** Lightweight Zustand stores (`authStore` , `authModalStore` , `toastStore` , `billingStore` , `limitStore`).
- **Data Visualization:** Recharts (Area charts for revenue trends, Donut charts for cost allocation).
- **Performance Optimization:** Below-the-fold dynamic code splitting (`next/dynamic`), GPU-accelerated canvas spotlights, Node `--max-old-space-size=8192` build booster.

⚙️ Backend Architecture (Microservices Layer)

- **Runtime:** Java 21 & Spring Boot 3.x.
 - **Service Breakdown:**
 - `api-gateway` : Central entry point handling JWT token filtering, CORS headers, and rate-limiting.
 - `auth-service` : Authentication, Registration, Google OAuth2, WhatsApp 6-digit OTP, Tenant Isolation, and Subscription Billing.
 - `crm-service` : Client lead pipeline Kanban board, Quotes, Proposals, and Vendor Agreements.
 - `event-service` : Run-of-Show stage scheduling, Crew dispatch, and Venue logistics.
 - `gallery-service` : AI Photo Booth media uploads, EXIF metadata extraction, and Cloudinary CDN storage.
 - **JVM Footprint Tuning:** Capped at `-Xmx128m` per microservice so all 5 backend services + Next.js frontend execute simultaneously under **1.2 GB RAM total**.
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3. Multi-Tenant Data Isolation & Security Model

Tenant Isolation Guarantee

- Every agency workspace is assigned a unique `tenantId` (e.g. `tenant_apex` , `tenant_royal` , `tenant_subhub`).
- Every HTTP request passes an `X-Tenant-Id` header validated against the user's JWT authorization claims.
- PostgreSQL database queries enforce `WHERE tenant_id = :tenantId` scoping, guaranteeing **100% data privacy** between competing event agencies.

Multi-Workspace Switcher (`WorkspaceSelectorPill.tsx`)

- Agency owners owning multiple companies can switch between active workspace contexts in 1 second without re-logging in.

Authentication Safeguards

- **1-Click Returning User Profile**: Recognizes returning browser sessions for 1-second sign-in ("Welcome back, Lokesh!").
 - **Email Domain Auto-Suggestion**: Suggests domain completions (`name@gma` → `name@gmail.com`) to eliminate signup typos.
 - **WhatsApp 6-Digit OTP**: Passwordless authentication for event clients.
 - **Logout Confirmation Popup Modal (`LogoutConfirmationModal.tsx`)**: High-contrast obsidian warning dialog protecting users from accidental sign-outs.
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4. Financial Engine & Monetization Theory

Tiered Subscription Plans

1. **STARTER**: Up to 3 Active Events, 5GB Storage, 1,000 AI Credits.
2. **PRO**: Unlimited Events, 50GB Storage, Custom Domain, White-Label Email Sender, Priority WhatsApp Dispatch.
3. **ENTERPRISE**: Custom Limits, Dedicated Database Shards, SLA Guarantee, 24/7 Phone Support.

Real-Time Financial Analytics & Margins

(`EventFinancialAnalytics.tsx`)

- Audits monthly gross revenue, vendor payouts, and net profit margins (`41.9% Margin`).
- Categorizes production expenses (Stage Decor 35%, Catering 28%, Sound & AV 20%, Cinematography 10%, Crew 7%).
- Per-Event Profitability Audit Table calculating net profit per contract.
- 1-Click Currency Toggle (`₹ INR` , `$ USD` , `€ EUR`).

Dynamic Payment Processing (`DynamicUpiQrModal.tsx` & Stripe)

- Integrated Stripe Checkout (`pk_live_...` / `sk_live_...`) for global credit card billing.
 - Real-Time Dynamic UPI QR Code modal with VPA verification (`eventos.pay@hdfcbank`) and 15-minute transaction expiration countdown.
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5. Complete App Route Taxonomy (48 Pages)

Category	Routes / Pages	Purpose
Public Landing	<code>/</code> , <code>/about</code> , <code>/features</code> , <code>/pricing</code> , <code>/calculator</code> , <code>/contact</code> , <code>/blog</code> , <code>/docs</code> , <code>/security</code> , <code>/status</code> , <code>/sla</code> , <code>/terms</code> , <code>/privacy</code> , <code>/cookies</code>	Marketing, ROI Calculator, Security & Legal disclosures
Authentication	<code>/login</code> , <code>/register</code> , <code>/forgot-password</code> , <code>/reset-password</code> , <code>AuthModal.tsx</code>	Password, Magic Link, Google OAuth, & WhatsApp 6-digit OTP Auth
Workspace & Onboarding	<code>/workspace-select</code> , <code>/onboarding</code> , <code>/accept-invite</code>	Multi-tenant workspace switcher & wizard setup
Quote Calculator	<code>/quote-calculator</code>	Interactive event budget estimator & 1-Click PDF proposal generator
Dashboard Core	<code>/dashboard</code> , <code>/ai</code> , <code>/chat</code> , <code>/activity</code>	Main workspace dashboard, AI Assistant, Team Chat, & Audit Logs
Operations	<code>/crm</code> , <code>/events</code> , <code>/bookings</code> , <code>/gallery</code>	Lead Kanban board, Calendar & Run-of-Show, Bookings, & Photo Booth
Finance	<code>/finance</code> , <code>/quotes</code> , <code>/payments</code> , <code>/invoices</code> , <code>/calculator</code>	Financial Analytics, Invoices, UPI QR Modal, & Payments Ledger
Intelligence	<code>/reports</code> , <code>/automation</code> , <code>/import</code>	Analytics graphs, Smart Automation Trigger builder, & CSV importer
Settings & Admin	<code>/settings</code> , <code>/settings/security</code> , <code>/portal</code> , <code>/developer</code> , <code>/superadmin</code>	White-label domain setup, Payment Gateways, WhatsApp API, & Client Portal



6. Summary Statement

EventOS is built from the ground up as a state-of-the-art, enterprise-grade SaaS operating system. It combines modern UI aesthetics (Linear/Vercel obsidian dark design), resilient microservices architecture (Java 21 / Spring Boot), robust multi-tenant data isolation, and comprehensive financial tools to power the next generation of event management agencies worldwide.